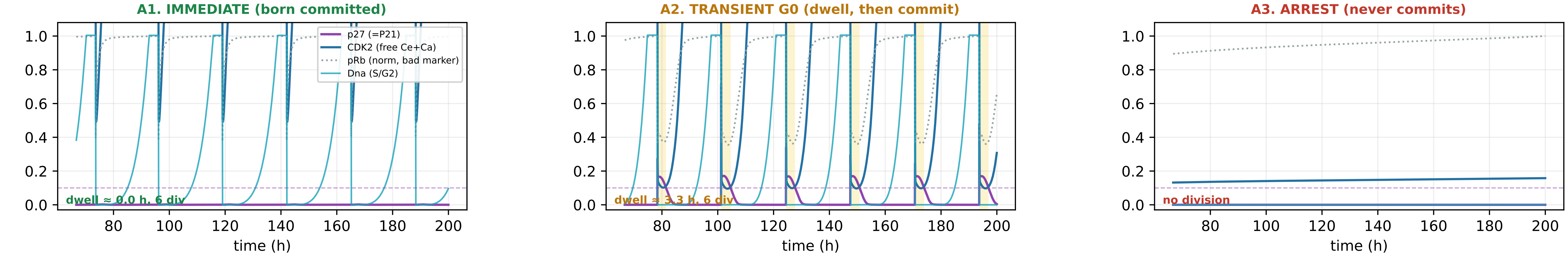
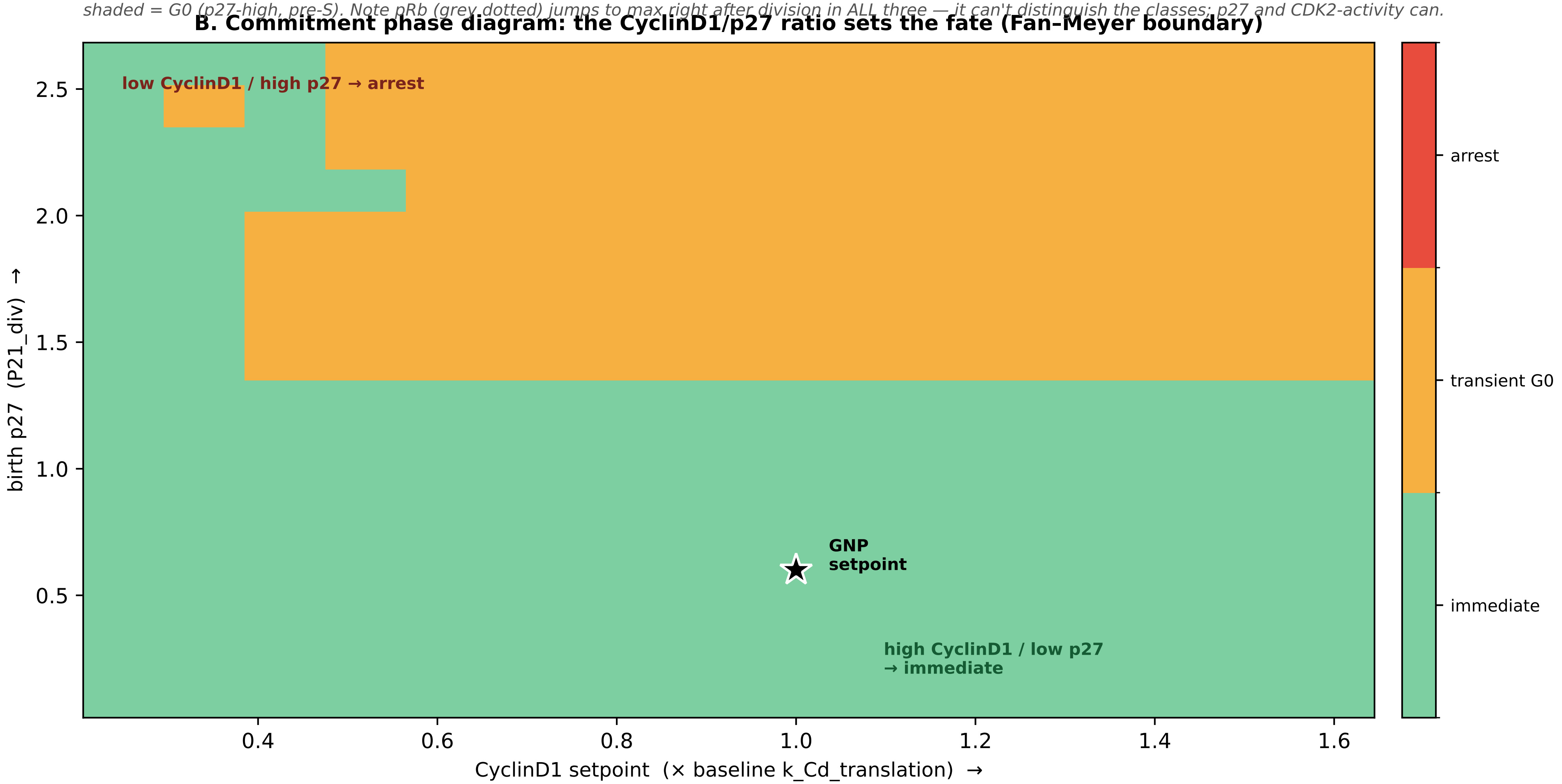
Transient G0 vs immediate G1/early-S entry — definition, mechanism, and CyclinD1/p27 control (v44)

G0 = p27-high / pre-S dwell after birth (NOT pRb, which saturates early). immediate: dwell ≤ 2 h · transient G0: commits after a dwell · arrest: never commits.



shaded = G0 (p27-high, pre-S). Note pRb (grey dotted) jumps to max right after division in ALL three — it can't distinguish the classes; p27 and CDK2-activity can.



DEFINITION (per daughter, at birth):

```
born-state = CyclinD1/p27 ratio
  (birth p27 = P21_div; CyclinD1 setpoint)

G0 dwell = time from birth until p27
  drops below threshold (Skp2–p27 toggle
  flips; CDK2 freed). Marked by p27, not pRb.

• IMMEDIATE  dwell ≤ 2 h
   (Spencer carryover keeps pRb/CycE/low-p27)

• TRANSIENT G0  dwells, then commits
   (grows to M_commit / CyclinD1 rises)

• ARREST  never commits
   (CyclinD1 below the threshold)
```